WORKED SOLUTION

STEP III 1988, Question 7

Pure • Differential Equations and Orthogonality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 8.6, *The Quantum Oscillator*: https://beyondstep.keithmcnulty.org/ch08/quantum-oscillator.html

The key idea

The first part is pure substitution: differentiate the trial functions twice and compare coefficients of powers of x, which pins down λ. The second part is the classic "Wronskian trick": the combination $y_m y_n' - y_n y_m'$ has a derivative in which the awkward $\omega^2 x^2$ terms cancel, leaving only a multiple of $y_m y_n$. Integrating that derivative over the whole line is then easy, because the conditions at $\pm\infty$ kill the boundary terms.

Solution

Throughout, primes denote $\mathrm{d}/\mathrm{d}x$. It helps to write the equation as

$$y_n'' = \left(\omega^2 x^2 - (2n+1)\omega\right) y_n. \qquad (*)$$

The case $n = 0$

With $y_0 = \mathrm{e}^{-\lambda x^2}$,

$$y_0' = -2\lambda x\, \mathrm{e}^{-\lambda x^2}, \qquad y_0'' = (4\lambda^2 x^2 - 2\lambda)\, \mathrm{e}^{-\lambda x^2}.$$

Substituting into $(*)$ with $n = 0$ and cancelling the non-zero factor $\mathrm{e}^{-\lambda x^2}$, we need

$$4\lambda^2 x^2 - 2\lambda = \omega^2 x^2 - \omega \quad \text{for all } x.$$

Comparing constant terms gives $\lambda = \frac{1}{2}\omega$, and then the x^2 terms agree too, since $4\lambda^2 = \omega^2$. (The other root $\lambda = -\frac{1}{2}\omega$ of $4\lambda^2 = \omega^2$ fails the constant term.)

The case $n = 1$

With $y_1 = x\,\mathrm{e}^{-\lambda x^2}$,

$$y_1' = (1 - 2\lambda x^2)\,\mathrm{e}^{-\lambda x^2},$$
$$y_1'' = \big(-4\lambda x - 2\lambda x(1 - 2\lambda x^2)\big)\mathrm{e}^{-\lambda x^2} = (4\lambda^2 x^3 - 6\lambda x)\,\mathrm{e}^{-\lambda x^2}.$$

Equation $(*)$ with $n = 1$ requires $y_1'' = (\omega^2 x^2 - 3\omega)\,x\,\mathrm{e}^{-\lambda x^2}$, that is

$$4\lambda^2 x^3 - 6\lambda x = \omega^2 x^3 - 3\omega x \quad \text{for all } x,$$

which again holds exactly when $\lambda = \frac{1}{2}\omega$. So the *same* constant works for both functions:

$$\lambda = \tfrac{1}{2}\omega, \qquad y_0 = \mathrm{e}^{-\omega x^2/2}, \qquad y_1 = x\,\mathrm{e}^{-\omega x^2/2}.$$

The conditions at infinity

Since $\lambda = \frac{1}{2}\omega > 0$, every function of the form (polynomial)$\times \mathrm{e}^{-\lambda x^2}$ tends to 0 as $x \to \pm\infty$, because $\mathrm{e}^{-\lambda x^2}$ decays faster than any power of x grows. The functions y_0, y_0', y_1 and y_1' are all of this form (the polynomials being 1, $-2\lambda x$, x and $1 - 2\lambda x^2$), so all four tend to 0 at both ends, as required.

The identity

By the product rule, the terms $y_m' y_n'$ cancel:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(y_m y_n' - y_n y_m'\big) = y_m' y_n' + y_m y_n'' - y_n' y_m' - y_n y_m'' = y_m y_n'' - y_n y_m''.$$

Now use $(*)$ for both y_n'' and y_m'':

$$\begin{aligned} y_m y_n'' - y_n y_m'' &= \big(\omega^2 x^2 - (2n+1)\omega\big) y_m y_n - \big(\omega^2 x^2 - (2m+1)\omega\big) y_m y_n \\ &= \big((2m+1) - (2n+1)\big)\omega\, y_m y_n \\ &= 2(m-n)\,\omega\, y_m y_n, \end{aligned}$$

which is the required result.

The orthogonality

> **Ask yourself** How do we know the infinite integral even exists, before we try to show it is zero?

We work on a finite interval first and let the ends go to infinity separately; existence then comes for free. Write $W = y_m y_n' - y_n y_m'$. For any $S, R > 0$, integrating the identity from $-S$ to R gives

$$2(m-n)\,\omega \int_{-S}^{R} y_m y_n \,\mathrm{d}x = W(R) - W(-S).$$

As $R \to \infty$, each of y_m, y_n, y_m', y_n' tends to 0 by the given conditions, so $W(R) \to 0$. Likewise $W(-S) \to 0$ as $S \to \infty$. Hence the right-hand side tends to 0, so the integral $\int_{-\infty}^{\infty} y_m y_n \,\mathrm{d}x$ converges and

$$2(m-n)\,\omega \int_{-\infty}^{\infty} y_m y_n \,\mathrm{d}x = 0.$$

If $m \neq n$ then $2(m-n)\omega \neq 0$, because $\omega > 0$. Dividing by it gives

$$\int_{-\infty}^{\infty} y_m(x)\, y_n(x) \,\mathrm{d}x = 0,$$

as required. Notice that this needs only the equation and the end conditions: we never needed formulas for y_m or y_n.

Checks

- For $m = 0$, $n = 1$ the result is obvious directly: $y_0 y_1 = x\,\mathrm{e}^{-\omega x^2}$ is odd, so its integral over the whole line is 0.
- A less trivial test: $y_2 = (2\omega x^2 - 1)\mathrm{e}^{-\omega x^2/2}$ satisfies the equation with $n = 2$ (try it). Using $\int_{-\infty}^{\infty} \mathrm{e}^{-\omega x^2}\,\mathrm{d}x = \sqrt{\pi/\omega}$ and $\int_{-\infty}^{\infty} x^2 \mathrm{e}^{-\omega x^2}\,\mathrm{d}x = \frac{1}{2\omega}\sqrt{\pi/\omega}$, we get $\int y_0 y_2 \,\mathrm{d}x = 2\omega \cdot \frac{1}{2\omega}\sqrt{\pi/\omega} - \sqrt{\pi/\omega} = 0$.
- Numerically, with $\lambda = \frac{1}{2}\omega$ the equation's residual at random points was below 10^{-6} for $n = 0, 1, 2$; the derivative identity held to 10^{-4} by finite differences; and Simpson's rule on $[-30, 30]$ gave all of $\int y_0 y_1$, $\int y_0 y_2$, $\int y_1 y_2$ smaller than 10^{-15} for $\omega = 0.5, 1, 2.5$.

Common slips

- Matching only the x^2 coefficients and getting $\lambda = \pm\frac{1}{2}\omega$: the constant term rules out the negative sign (and so does the decay at infinity).
- Forgetting to say why the boundary terms vanish: this is exactly where both conditions, on y_n *and* on y_n', are used.
- Dividing by $m - n$ without noting that it is non-zero and that $\omega > 0$.

Beyond STEP

This is the quantum harmonic oscillator: the y_n are its energy states, $(2n+1)\omega$ are (up to scaling) its energy levels, and the functions are Hermite polynomials times a Gaussian. The orthogonality argument is a special case of Sturm–Liouville theory, which shows that eigenfunctions of a self-adjoint differential operator with different eigenvalues are always orthogonal, just as eigenvectors of a symmetric matrix are. Boyce and DiPrima, *Elementary Differential Equations*, covers boundary-value problems and Sturm–Liouville theory.

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